Lecture 18

Saturday, April 18, 2026 9:24 AM

Recap: GDA (classification technique)

- Models $P(x|y)$, $P(y)$ (x: feature vector, y: class)
- Assumes $x|y=k \sim \mathcal{N}(\mu_k, \Sigma)$, $y \sim Ber(\phi)$
- Parameters: $\phi, \mu_0, \mu_1, \Sigma$
- Uses MLE to estimate parameters (closed form, $O(Nd^2)$ complexity)
- Prediction: $\hat{y} = \arg\max_y \mathbb{P}(y|x)$ ←

$$= \arg\max_y \mathbb{P}(x|y)\,\mathbb{P}(y) \quad \text{(Bayes' rule)}$$

Naïve Bayes

Limitations of GDA

Spam filter (spam / not spam)

$d = 50000$

set of 50000 "important" words: buy, free, offer, meeting, ...

$$x_j = \begin{cases} 1 & \text{if word } j \text{ occurs in the email} \\ 0 & \text{otherwise} \end{cases}$$

GDA on this:

$\mu_k \quad k \in \{0,1\} \rightarrow 50000 \quad (100,000)$

$\Sigma \rightarrow 50000 \times 50000 : 25 \times 10^8$

(Need something else to make this tractable)

- Naïve Bayes (conditional independence)

It models $\mathbb{P}(x|y)$ & $\mathbb{P}(y)$

Assumes:

$$\mathbb{P}(x_1, x_2, \dots, x_d \mid y) = \prod_{j=1}^{d} \mathbb{P}(x_j|y)$$

Given the class, the features are independent

$d = 3$ (say)

$$\mathbb{P}(x_1, x_2, x_3 \mid y) = \mathbb{P}(x_1|y)\,\mathbb{P}(x_2|y)\,\mathbb{P}(x_3|y)$$

In reality:

$$\mathbb{P}(x_1, x_2, x_3 \mid y) = \mathbb{P}(x_1|y)\,\boxed{\underbrace{\mathbb{P}(x_2 \mid y, x_1)}_{\substack{\shortparallel \\ \mathbb{P}(x_2|y)}}}\,\underbrace{\mathbb{P}(x_3 \mid y, x_1, x_2)}_{\substack{\shortparallel \\ \mathbb{P}(x_3|y)}}$$

$x_2 =$ "free"
$x_3 =$ "offer"

$$\mathbb{P}(x_2 \mid y, x_3) = \mathbb{P}(x_2 \mid y) \leftarrow \text{Naïve Bayes}$$

K class problem
$x_i \in \{0,1\}$, d-dimensional features

$p(y=k) \quad k = 0, 1, \ldots, K-1 \quad$ (K parameters)
$\mathbb{P}(x_i \mid y=k) \quad d \times k \quad$ (dK parameters)

GDA : $O(d^2)$
Naive Bayes : $\cancel{K+1}$ $(d+1)K$

Naïve Bayes Assumptions:

1. $\mathbb{P}(y=k) = \phi_K \qquad \sum_{k=0}^{K} \phi_K = 1$
2. $\mathbb{P}(x \mid y=k) = \prod_{j=1}^{d} \mathbb{P}(x_j \mid y=k)$ (conditional independence)
3. $x_j \in \{0,1\}$

$$x_j \mid y=k \sim \text{Ber}(\phi_{j|k})$$

j ↓ feature index; k ↳ class label

Find $\phi_{j|k}$'s, ϕ_K's $\quad (d+1)K$ parameters

1. How to estimate the parameters? ✓
2. How to make a prediction for test samples?

↳ Training data: $\{x^{(i)}, y^{(i)}\}_{i=1}^{N}$

$$L = \prod_{i=1}^{N} \mathbb{P}(x^{(i)}, y^{(i)}) \quad \text{— ①}$$

$$l = \log L = \sum_{i=1}^{N} \log \underbrace{\mathbb{P}(x^{(i)}, y^{(i)})} \quad \text{— ②}$$

$$= \sum_{i=1}^{N} \log \underbrace{\mathbb{P}(x^{(i)} | y^{(i)})} \mathbb{P}(y^{(i)}) \quad \text{— ③}$$

NB assumption

$$= \sum_{i=1}^{N} \log \underbrace{\prod_{j=1}^{d} \mathbb{P}(x_j^{(i)} | y^{(i)}) \; \mathbb{P}(y^{(i)})} \quad \text{— ④}$$

$$= \underbrace{\sum_{i=1}^{N}}_{\text{training sample}} \Big(\underbrace{\log \mathbb{P}(y^{(i)})}_{\text{class}} + \underbrace{\sum_{j=1}^{d} \log \mathbb{P}(x_j^{(i)} | y^{(i)})}_{\text{features, class}} \Big) \quad \text{⑤}$$

Solution: (closed form, easy to interpret)

① $\phi_k^* = \mathbb{P}(y=k) = \dfrac{\text{count}(y^{(i)}=k)}{N}$ (fraction of class k samples in the training data)

$k \in \{0, 1, \ldots K-1\}$

② $\underbrace{\phi^*_{j|k}} = \dfrac{\text{count}(x_j^{(i)} = 1 \text{ and } y^{(i)} = k)}{\text{count}(y^{(i)} = k)}$ ← $\quad k \in \{0,1,\ldots K-1\}$, $\underline{\underline{j}} \in \{1, 2, \ldots d\}$

Probability $x_j = 1$ given class is k

Going to back spam email example:

N − # of emails, ϕ_k^* = fraction of spam emails

j − feature index (corresponds to a word)

k − class index

O(Nd) computation

Q. How to make a prediction?

$$\hat{y} = \underset{y}{\arg\max}\; \mathbb{P}(y=k | x)$$

feature vector of test sample

$$= \underset{y}{\operatorname{argmax}} \; \mathbb{P}(x \mid y = k)\, \mathbb{P}(y = k)$$

$$= \underset{y}{\operatorname{argmax}} \; \prod_{j=1}^{d} \underbrace{\mathbb{P}(x_j \mid y = k)}_{= \phi^*_{j|k}} \; \underbrace{\mathbb{P}(y = k)}_{= \phi^*_k}$$

We saw an issue with multiplications by 0.
We need a fix.

Laplace Smoothening

- We pretend we saw every word at least once in each class.

$$\hat{\phi}_{j|k} = \frac{\text{count}(x_j = 1 \;\&\; \text{class } k)}{\text{count}(\text{class } k)} \quad \Big\} \text{ original ✗}$$

$$= \frac{\text{count}(x_j = 1 \;\&\; \text{class } k) + 1}{\text{count}(\text{class}) + \text{number of values } x_j \text{ can take}} \quad \Big\} \text{ new ✓}$$

$$\hat{\phi}_{j|k}\ (\text{new}) > 0$$

Question:

independence

Case I: $A \perp\!\!\!\perp B \Rightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$

$\not\Uparrow \qquad \not\Downarrow$

Case II: $A \perp\!\!\!\perp B \mid C \Rightarrow \mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\,\mathbb{P}(B \mid C)$

conditionally independence

All models are wrong; but some are useful